

THE SIGNALRUPTURE STRUCTURAL GRAMMAR
EMPIRICAL VALIDATION I: CANADIAN HEALTHCARE
Requirement, Durable Capacity, Compensation, and Function Under Persistent
Workforce Pressure

SignalRupture (SR)

Final Historical Empirical Paper — August 2026

Abstract

This paper conducts the first consolidated empirical test of the frozen SignalRupture (SR) Structural Grammar in Canadian healthcare. Rather than treating high pressure, high overtime, or poor access as sufficient evidence of structural failure, the test evaluates a narrower relational architecture: effective requirement (Q), durable capacity (D), sustainable flexibility (C_S), extraordinary compensation (C_E), and observable function (F). The central question is whether Canadian healthcare exhibits the sequence predicted by the grammar: requirement exceeding durable capacity, compensatory expansion preserving substantial function, and either restoration or compensatory constraint followed by deterioration. The analysis integrates Canadian Institute for Health Information (CIHI) hospital staffing data, CIHI hospital-harm and service-volume indicators, CIHI surgical and diagnostic wait-time evidence, emergency-department access evidence, and Statistics Canada health-occupation vacancy data.

The evidence establishes a strong descriptive requirement–capacity mismatch. Statistics Canada reports that employment in health-related occupations expanded over the long run, yet the health-occupation vacancy rate increased from 2.1% in 2016 to 5.8% in 2024; registered-nurse vacancy rates increased from 2.6% to 7.0%, licensed-practical-nurse vacancy rates from 2.5% to 12.8%, and personal-support-worker vacancy rates from 2.6% to 6.7%. In hospitals, CIHI reports that overtime hours approximately doubled and purchased staffing hours increased 126% between 2019–2020 and 2023–2024. The reconstructed overtime-plus-purchased share rose from approximately 3.24% to 5.88%, peaking near 6.08% in 2022–2023. Rural and remote hospitals show the strongest purchased-staffing dependence: purchased hours rose 261% over the period and represented more than 3% of worked hours in 2023–2024, compared with less than 1% in urban hospitals.

Function did not disappear. Acute inpatient hospitalizations reached 3.05 million in 2023–2024, up from 2.96 million the prior year. Procedure volumes also increased materially between 2019 and 2024. Yet access did not normalize proportionally: only 68% of hip replacements and 61% of knee replacements were completed within the 26-week benchmark in 2024, below 75% and 70% respectively in 2019; median MRI waits were 15 days longer; and several cancer-surgery waits increased. In 2024–2025, 16.1 million emergency-department visits were reported to CIHI, while 1.2 million visits—7.7%—ended before physician assessment. Hospital Harm rose from the pre-pandemic range of approximately 5.3–5.4% to roughly 6% and then plateaued.

The combined evidence supports two SR propositions: Surface–Substrate Divergence and descriptive Compensated Continuity. It does not establish Compensatory Inversion. The slight aggregate decline in compensation share in 2023–2024 coincided with continued growth in purchased staffing and no further increase in national Hospital Harm. This is correctly classified as indeterminate rather than exhaustion. The evidence also reveals the principal statistical limitation: public national series do not provide a matched hospital-level longitudinal panel linking requirement, durable staffing, extraordinary compensation, and subsequent function. Consequently, this paper does not claim Level-2 predictive validation. It establishes a stronger empirical baseline: the frozen grammar is compatible with observed Canadian healthcare dynamics, survives a meaningful non-confirmatory case, and generates a precisely specified next test capable of falsifying the predictive claim.

Keywords: SignalRupture; structural grammar; Canadian healthcare; workforce vacancies; overtime; agency staffing; hospital harm; wait times; compensated continuity; predictive validation.

1. Research Question

The empirical question is not whether Canadian healthcare is “under strain.” That proposition is already well documented. The question is whether the strain can be represented prospectively through the frozen SR grammar without changing the meaning of its variables after observing the data.

$$Q \leftrightarrow D \leftrightarrow C_S / C_E \leftrightarrow F$$

The test asks four linked questions. First, is there evidence that effective requirement has outgrown durable capacity? Second, has compensatory capacity expanded in response?

Third, has substantial function been preserved despite the altered substrate? Fourth, when compensation ceases to expand, does function subsequently deteriorate—or does the evidence instead indicate restoration, stabilization, or an indeterminate state?

2. Frozen SR Hypotheses

ID	Prediction	Status before this test
P1	Persistent Q>D plus expandable C_E preserves F relative to otherwise comparable low-compensation systems.	Prospective
P2	Similar or substantial F can coexist with materially different substrate sufficiency and compensation dependence.	Prospective with descriptive support
P3	Persistent Q>D plus plateauing/falling C_E predicts subsequent deterioration in F.	Prospective
P4	Closing Q–D permits C_E to fall while F remains stable or improves.	Prospective
P5	The same relational roles should translate to heterogeneous domains without post-hoc remapping.	Operationally supported elsewhere

3. Measurement Architecture

SR role	Healthcare operationalization	Evidence used
Q	Demand/service requirement: population, acuity, procedures, ED visits, staffing need, vacancies as unmet labour demand.	Statistics Canada; CIHI
D	Regular sustainable staffing and durable workforce capacity.	CIHI regular hours; workforce employment
C_S	Normal renewable flexibility, including ordinary scheduling and baseline overtime.	Conceptual boundary; not separately observed in aggregate data
C_E	Overtime above normal conditions, purchased/agency staffing, emergency redeployment.	CIHI overtime and purchased hours
F	Service volume, access, wait-time performance, patient safety, ED completion.	CIHI hospitalizations, waits, ED, Hospital Harm

$$G_C = Q - D$$

Because Q and D are multidimensional, this paper does not manufacture a universal scalar gap. The empirical burden is instead triangulated: durable staffing grows, yet vacancies and service requirements remain elevated; extraordinary staffing rises; and access outcomes remain impaired. This is stronger than treating spending or headcount alone as capacity.

4. Data and Source Discipline

All empirical claims in this paper are drawn from official Canadian sources. Hospital staffing hours are from CIHI’s 2024 health-workforce staffing analysis. CIHI notes that

the hospital staffing series excludes Quebec and Nunavut, excludes Saskatchewan in 2022–2023 and 2023–2024, and excludes Nova Scotia purchased hours in 2023–2024. These coverage changes matter and prevent interpretation of the aggregate series as a complete national census (CIHI, 2025a).

Vacancy evidence is drawn from Statistics Canada’s Job Vacancy and Wage Survey and Labour Force Survey analysis. The series provides annual occupation-specific vacancy rates from 2016 through 2024 and regional vacancy rates for 2024 (Statistics Canada, 2025). Functional outcomes are drawn from CIHI’s Hospital Harm indicator, hospital-stay statistics, priority-procedure wait-time analyses, and 2026 emergency-department analysis (CIHI, 2025b, 2025c, 2025d, 2026).

5. Requirement–Capacity Evidence: Workforce Vacancies

The strongest long-run evidence that durable workforce expansion has not fully closed requirement is the divergence between employment growth and vacancy pressure. Statistics Canada reports 1.34 million people working in health-related occupations in 2024. Employment in these occupations grew 71% from 2004 to 2024, and the health workforce’s average annual growth rate exceeded that of employment overall. Yet health-related occupations represented 15.0% of all Canadian job vacancies in 2024, up from 6.5% in 2016. The health-occupation vacancy rate rose from 2.1% to 5.8% over the same period (Statistics Canada, 2025).

Year	RN/RPN/NP vacancy rate	LPN vacancy rate	PSW vacancy rate
2016	2.6%	2.5%	2.6%
2017	2.7%	4.2%	3.3%
2018	3.4%	4.3%	4.2%
2019	3.9%	5.0%	4.8%
2020	5.8%	8.0%	7.2%
2021	6.8%	12.7%	7.8%
2022	7.2%	15.5%	8.5%
2023	8.0%	18.7%	8.2%
2024	7.0%	12.8%	6.7%

The 2024 decline from 2023 peaks is important. SR cannot describe every improvement as deterioration. Registered-nurse, LPN, and PSW vacancy rates all fell in 2024. Nevertheless, they remained substantially above 2016 levels, and Statistics Canada

concluded that unmet labour demand remained significant. The correct SR classification is persistent but partially easing requirement–capacity pressure, not monotonic collapse.

6. Regional Heterogeneity

National averages conceal materially different labour conditions. In 2024, the health-related occupation vacancy rate was 7.8% in the territories, 7.4% in Quebec, 6.2% in Atlantic Canada, 5.6% in Ontario, 5.5% in British Columbia, and 4.2% in the Prairies (Statistics Canada, 2025). Statistics Canada further reports that nursing vacancy rates were especially high in remote areas and that remote vacancies tended to remain open longer. This matters because the grammar predicts earlier compensation dependence where Q–D is larger and substitution is weaker.

$$RD_{region} \neq RD_{national}$$

7. Compensation Expansion in Hospitals

CIHI directly observes the compensatory staffing response. In 2023–2024, hospitals in the published series used more than 7.8 million purchased staffing hours, equivalent to 4,010 FTEs and 126% above 2019–2020. Providers also worked almost 32 million overtime hours, equivalent to 16,397 FTEs and approximately double 2019–2020 (CIHI, 2025a).

Fiscal year	Regular hours	Overtime hours	Purchased hours	Overtime + purchased share
2019–2020	576,604,708	15,824,632	3,467,567	3.24%
2020–2021	594,987,973	18,159,208	3,661,381	3.54%
2021–2022	602,552,932	26,258,321	4,820,469	4.90%
2022–2023	598,826,644	32,131,713	6,630,954	6.08%
2023–2024	636,691,376	31,973,251	7,820,301	5.88%

$$C_{share} = (Overtime + Purchased) / (Regular + Overtime + Purchased)$$

The reconstructed compensation share nearly doubled relative to its starting level. Regular hours rose approximately 10.4% over the interval, compared with approximately 102% growth in overtime and 125.5% growth in purchased staffing. This is a compositional shift, not merely an increase in the size of the hospital workforce.

8. Rural/Remote Compensation as a Natural Exposure Contrast

The urban–rural split supplies the strongest available quasi-comparison in the public staffing data. Purchased hours rose 261% in rural and remote hospitals from 2019–2020 to 2023–2024, versus 98% in urban hospitals. In 2023–2024, purchased staffing accounted for more than 3% of worked hours in rural/remote hospitals but less than 1% in urban hospitals. Overtime represented roughly 6% of all worked hours in rural/remote hospitals and 5% in urban hospitals in the broader staffing series (CIHI, 2025a).

A separate CIHI nursing-unit analysis reports an even higher overtime share among front-line providers: 8.0% nationally in 2023–2024, 9.2% in rural/remote hospitals and 7.9% in urban hospitals (CIHI, 2025e). Different denominators explain why these percentages should not be mechanically merged. The directional result is nevertheless stable: more exposed settings rely more heavily on compensatory staffing.

9. Function Was Preserved at Scale

The compensation period was not a period of simple cessation. CIHI reports 3.05 million acute inpatient hospitalizations in 2023–2024, up from 2.96 million in 2022–2023; age-adjusted average length of stay remained 7.3 days (CIHI, 2025c). This is central to the SR grammar. A system can preserve large service volumes while its staffing composition changes materially.

F_volume substantial $\wedge$ C_E elevated

10. But Volume Did Not Equal Restored Access

Priority-procedure evidence shows why output alone is insufficient. Between 2019 and 2024, procedure volumes rose: hip replacements increased 26%, knee replacements 21%, cataract surgeries 11%, cancer surgeries 7%, and MRI/CT scans 16%. Yet access benchmarks did not fully recover. In 2024, 68% of hip replacements were completed within 26 weeks versus 75% in 2019; knee replacements were 61% versus 70%; cataract surgery was approximately unchanged at 69% versus 70%. Radiation therapy within 28 days and hip-fracture repair within 48 hours each declined three percentage points from 2019. Median MRI waits increased by 15 days and CT waits by 3 days (CIHI, 2025d).

Volume $\uparrow$ does not imply Access restored

11. Emergency Department Function and Latency

CIHI's 2026 emergency-department analysis adds a current functional layer. In 2024–2025, 16.1 million ED visits were reported to CIHI's National Ambulatory Care Reporting System, representing about 89% of estimated Canadian ED visits. Two-thirds were triaged as high acuity. Half of patients waited just under two hours for initial physician assessment, while one in ten waited more than five hours. Approximately 1.2 million visits—7.7%—ended with the patient leaving before physician assessment (CIHI, 2026).

This is a particularly useful SR pattern: throughput remains enormous, but latency and incomplete access remain visible. It supports a distinction between function as volume and function as timely effective access.

12. Patient Safety: Hospital Harm

CIHI's Hospital Harm indicator measures acute-care hospitalizations with at least one potentially preventable harmful event. The indicator is available nationally, excluding Quebec, from 2014 through 2024. CIHI reported a pre-pandemic range around 5.3–5.4%, an increase to 5.9% in 2020–2021, and approximately 6.0% in subsequent years. In 2023–2024, roughly one in 17 hospital stays involved unintended harm (CIHI, 2025b; CIHI, 2024).

The plateau matters. It is inconsistent with a simplistic claim that rising compensation must produce continuously worsening harm. Under the frozen grammar, high compensation can be successful buffering. Hospital Harm therefore functions here as a negative control against overclaiming.

13. Testing P1 — Compensated Continuity

$G_C > 0 \wedge C_E \uparrow \rightarrow F$ deteriorates less than under comparable low- C_E conditions

The evidence is strongly compatible with P1 but does not causally confirm it. Workforce vacancies indicate unmet labour demand. Overtime and purchased staffing expanded sharply. Hospitals continued large-scale inpatient, surgical, imaging, and ED activity. CIHI explicitly describes overtime and purchased staffing as mechanisms hospitals use to fill staffing gaps and maintain care delivery (CIHI, 2025a, 2025e).

What is missing is the counterfactual. We do not yet possess a matched hospital panel showing that, conditional on similar Q–D gaps, hospitals with greater expandable C_E

preserved F better than hospitals with less C_E. P1 is therefore scored Partially Supported, not Confirmed.

14. Testing P2 — Surface–Substrate Divergence

$$F_A \approx F_B \text{ does not imply } D_A/Q_A \approx D_B/Q_B$$

P2 receives the strongest support. Service volumes recovered or increased in several areas while vacancy pressure, overtime, agency staffing, wait-time deficits, and ED latency remained elevated. The same surface quantity—procedures completed—can therefore coexist with different levels of staffing compensation and access performance. P2 is scored Supported descriptively.

15. Testing P3 — Compensatory Inversion

$$CI_t = [\Delta C_E \leq 0] \wedge [G_C > 0] \wedge [F_{(t+h)} \text{ deteriorates}]$$

The public aggregate data do not demonstrate P3. The combined overtime-plus-purchased share eased from approximately 6.08% in 2022–2023 to 5.88% in 2023–2024. But purchased staffing continued to increase, regular hours increased substantially, and national Hospital Harm remained approximately flat. Meanwhile, occupation vacancy rates eased in 2024 from 2023 peaks. These observations are compatible with stabilization, compositional substitution, partial restoration, or unresolved pressure; they are not a clean compensatory inversion.

This non-confirmation is important. The frozen protocol prevents SR from labeling any decline in overtime as exhaustion. P3 is scored Indeterminate / Not Observed in the national aggregate series.

16. Testing P4 — Restoration

$$G_C \downarrow \rightarrow C_E \text{ dependence } \downarrow \wedge F \text{ stable/improving}$$

There are hints of a restoration branch but not enough matched evidence to confirm it. Regular hospital hours rose markedly in 2023–2024 and several health-occupation vacancy rates declined in 2024. At the same time, compensation share eased slightly and Hospital Harm did not worsen. However, access indicators remained impaired, and the staffing and vacancy series are not matched at the same unit or exact time basis. The correct score is Indeterminate with a candidate restoration signal.

17. A Descriptive Timing Diagnostic

Using the five annual hospital staffing observations and approximate national Hospital Harm values, the same-year correlation between compensation share and harm is about 0.74 and the one-period lag correlation about 0.61. These values are not inferential evidence. The effective samples are five and four observations, respectively, and the COVID-19 period jointly changed demand, acuity, staffing availability, care mix, and patient risk. Reporting a p-value or fitting a predictive regression to these data would create false precision.

$$r_{\text{same-year}} \approx 0.74; \quad r_{\text{lag-1}} \approx 0.61 \quad [\text{descriptive only}]$$

18. Stronger Historical Evidence From Vacancies

The vacancy series provides nine annual observations and is more useful for establishing persistence of the requirement–capacity problem than the five-year hospital-hours series. All three frontline groups—registered nurses, LPNs, and PSWs—show a sharp post-2019 rise followed by partial easing in 2024. This pattern is consistent with a system in which requirement rose faster than immediately available durable labour capacity and later began to adjust. It does not by itself identify the contribution of compensation to functional outcomes.

The key result is therefore relational rather than causal: durable workforce growth and high employment did not eliminate unmet labour demand. This supports the SR insistence that gross capacity additions should be evaluated against changing requirement.

19. Why a National Aggregate Cannot Deliver Predictive Validation

- Q, D, C_E, and F are not observed at a common hospital-level unit.
- Hospital staffing coverage changes across years and jurisdictions.
- Hospital Harm is reported nationally in the public indicator, not matched to staffing exposure.
- Wait-time datasets vary by procedure and jurisdiction.
- Calendar-year vacancy data and fiscal-year staffing data are not perfectly aligned.
- COVID-19 creates a major common shock and structural break.

- Only five annual hospital staffing observations are available in the published aggregate series.
- Normal renewable overtime cannot be separated empirically from extraordinary overtime without a baseline model.
- National aggregation can hide local compensatory inversion and local restoration occurring simultaneously.

20. The Decisive Statistical Test

The next test should not be another national correlation. It requires a longitudinal multi-unit panel. Unit i should be a hospital, health region, hospital system, occupation, or service line; t should preferably be quarterly. The panel should contain a pre-period sufficient to estimate normal compensation and a later held-out period for genuine prediction.

$$M_0: F_{(i,t+h)} = f(F_{(i,t)}, Q_{(i,t)}, D_{(i,t)}, Z_{(i,t)})$$

$$M_{SR}: M_0 + C_{E,(i,t)} + \Delta C_{E,(i,t)} + [G_C(i,t) \times \Delta C_{E,(i,t)}]$$

$$\Delta \text{Performance} = \text{Performance}(M_{SR}) - \text{Performance}(M_0)$$

SR moves to domain predictive validity only if the SR augmentation improves out-of-sample forecast performance, calibration, warning lead time, or decision utility beyond a credible healthcare baseline.

21. Minimum Panel Specification

Component	Preferred measure
Unit	Hospital / health region / service line
Frequency	Quarterly; monthly where stable
Q	ED visits, admissions, acuity, referrals, procedure demand, occupancy, required staffing
D	Permanent/regular FTEs, staffed beds, regular worked hours, durable recruitment
C_S	Historical normal overtime, float pools, casual/part-time renewable flexibility
C_E	Overtime above baseline, agency/purchased hours, emergency redeployment
F-access	Wait distributions, benchmark attainment, left-without-being-seen, closures/diversions
F-quality	Hospital Harm or matched safety indicators
F-workforce	Absence, turnover, vacancy duration, retention
Controls	Seasonality, COVID burden, demographics, acuity/case mix, funding, wage/policy changes, geography

22. Identification Strategy

A credible design should exploit variation in compensation exposure across regions and hospitals rather than infer from national time trends. Rural/remote versus urban variation is one candidate exposure axis, but it must be matched to outcomes and confounders.

Province-specific policy changes, agency-staffing restrictions, wage settlements, training expansions, and recruitment programs may also provide quasi-experimental variation if their timing and assignment permit defensible identification.

For P1, the target is the buffering effect of compensation under comparable gaps. For P3, the target is whether a compensation plateau or decline under an unresolved gap predicts later deterioration. For P4, the target is whether durable capacity expansion reduces extraordinary compensation without harming function. These are distinct estimands and should not be collapsed into one coefficient.

23. Statistical Validation Rules

- Freeze outcome definitions and horizons before the holdout period is examined.
- Estimate normal C_S from a pre-specified historical baseline; classify only excess/non-renewable use as C_E.
- Use hospital or region fixed effects where appropriate to absorb stable structural differences.
- Model seasonality and major policy/COVID structural breaks explicitly.
- Use clustered or otherwise appropriate uncertainty estimates for repeated observations.
- Report calibration and forecast error, not significance alone.
- Compare against strong healthcare baselines rather than an intentionally weak model.
- Run sensitivity analyses across reasonable Q definitions.
- Score restoration cases as positively as deterioration cases.
- Publish null and falsifying results.

24. Empirical Scorecard

Prediction	Evidence	Final score
P1 Compensated Continuity	Vacancies high; overtime/agency use expanded; substantial service continued; no matched counterfactual.	Partially Supported
P2 Surface–Substrate Divergence	High/increasing volumes coexist with elevated compensation, vacancies and impaired access.	Supported descriptively

P3 Compensatory Inversion	No clean aggregate inversion; final compensation-share dip not followed by worse harm.	Indeterminate / Not observed
P4 Restoration	Regular hours rose and vacancies eased, but access remained strained and data are unmatched.	Indeterminate; candidate signal
Healthcare domain predictive validity	No matched longitudinal out-of-sample test yet.	Not achieved

25. What This Paper Confirms

- Canadian healthcare can be operationalized using the frozen Q–D–C–F grammar without changing the roles after observing outcomes.
- Unmet labour demand persisted despite substantial growth in health employment and hospital staffing.
- Compensatory staffing expanded much faster than regular hospital hours from 2019–2020 to 2023–2024.
- Large-scale service output can coexist with elevated compensation dependence and impaired access.
- Rural/remote systems exhibit stronger purchased-staffing dependence, consistent with heterogeneous gap exposure.
- High compensation alone does not imply imminent deterioration; Hospital Harm plateaued while compensation remained elevated.

26. What This Paper Does Not Confirm

- It does not establish that overtime or agency staffing caused changes in patient outcomes.
- It does not establish a national Compensatory Inversion.
- It does not prove a scalar Q–D gap.
- It does not show that compensation predicts later function beyond established healthcare indicators.
- It does not establish Level-2 predictive validity.
- It does not establish predictive metatheoretical validation.

27. Implications for the Structural Grammar

The healthcare evidence strengthens SR by narrowing it. The grammar survives not because every indicator worsens, but because it correctly allows multiple simultaneous

states: persistent requirement pressure, compensatory expansion, preserved output, impaired access, partial workforce recovery, and stable patient-safety rates. A theory that required monotonic deterioration would be contradicted by the data. The frozen SR grammar does not require that pattern.

The empirical contribution at this stage is diagnostic decomposition. Canadian healthcare demonstrates why output, capacity, compensation, and access cannot be treated as interchangeable measures. Whether that decomposition becomes predictively superior remains an open empirical question.

28. Relationship to the Prospective Prediction

This historical paper must remain separate from the August 2026 prospective Structural Grammar Prediction. The historical evidence can inform measurement, but it cannot rewrite the frozen future hypotheses. The prospective prediction remains that unresolved $Q > D$ with expandable compensation should preserve function temporarily; restoration should reduce extraordinary dependence without worsening function; and constrained compensation under an unresolved gap should precede functional deterioration.

Historical test informs measurement; future test scores prediction.

29. Final Empirical Judgment

Canadian healthcare provides the strongest empirical domain yet for the SR Structural Grammar, but the result is not predictive validation. The evidence establishes persistent labour-demand pressure, extraordinary compensatory expansion, continued large-scale function, and incomplete access recovery. It also provides a useful non-confirmation: the aggregate series does not show the frozen Compensatory Inversion pattern. That restraint is scientifically important.

Operational fit: PASS

Descriptive compensated continuity: PARTIAL SUPPORT

Surface–substrate divergence: SUPPORT

Compensatory inversion: NOT OBSERVED / INDETERMINATE

Domain predictive validity: NOT YET ACHIEVED

30. Conclusion

The Canadian healthcare test demonstrates why SR should be evaluated as a structural grammar rather than as a vocabulary of crisis. Between 2016 and 2024, health-workforce employment expanded while vacancy pressure rose substantially. Between 2019–2020 and 2023–2024, hospital overtime and purchased staffing expanded far faster than regular hours. Hospitals nevertheless continued to deliver millions of inpatient stays, procedures, diagnostic tests, and emergency visits. At the same time, several access indicators remained worse than their pre-pandemic benchmarks and patient-safety harm remained elevated relative to the pre-pandemic period.

This configuration is consistent with compensated continuity and strongly supports surface–substrate divergence. It is not evidence that Canadian healthcare is collapsing, and it is not evidence that high compensation necessarily signals exhaustion. Indeed, the absence of a clean aggregate compensatory inversion is a critical result: the frozen grammar survives because it distinguishes high compensation from constrained compensation and permits restoration or stabilization.

The empirical frontier is now clear. The next valid step is a matched longitudinal hospital or regional panel capable of estimating the requirement–capacity gap, distinguishing sustainable flexibility from extraordinary compensation, and testing whether compensation trajectory predicts future access, workforce, or quality outcomes beyond established healthcare models. Until that test is completed, SR remains at operational invariance with partial domain evidence rather than predictive validation.

The result is therefore neither a confirmation by rhetoric nor a failure. It is a bounded empirical finding: Canadian healthcare behaves in a way that makes the frozen SR grammar measurable, some of its diagnostic predictions are observed, its strongest prospective transition is not yet observed, and the data requirements for a decisive falsifiable test are now specified.

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Appendix A. Frozen Variable Register

Symbol	Meaning	Healthcare implementation
Q	Effective requirement	Demand, acuity, access targets, staffing requirement
D	Durable capacity	Regular staffing, permanent FTEs, staffed service capacity
C_S	Sustainable flexibility	Normal renewable scheduling/float/overtime
C_E	Extraordinary compensation	Excess overtime, agency/purchased staffing, emergency redeployment
F	Observable function	Volume, access, waits, safety, completion
G_C	Requirement-capacity gap	Q–D, measured as domain-specific vector
CI	Compensatory Inversion	C_E plateaus/falls under unresolved gap and later F worsens

Appendix B. Data-Coverage Warnings

- CIHI hospital staffing totals exclude Quebec and Nunavut.
- Saskatchewan is excluded from the staffing series in 2022–2023 and 2023–2024.
- Nova Scotia purchased hours are excluded in 2023–2024.
- CIHI Hospital Harm national reporting excludes Quebec.
- Emergency-department NACRS reporting represented about 89% of Canadian ED visits in 2024–2025.
- Vacancy series are calendar-year labour-market measures; hospital staffing series are fiscal-year measures.
- Different CIHI staffing analyses use different provider/setting denominators; percentages should not be merged mechanically.

Appendix C. Frozen Next-Test Protocol

Stage	Rule
Development	Define Q, D, C_S, C_E and F using pre-holdout data only.
Baseline	Fit best conventional healthcare forecast model M_0 .
SR augmentation	Add gap, compensation level, compensation trajectory and inversion state.
Holdout	Predict later hospitals/quarters not used in development.
Primary comparison	Out-of-sample error/calibration and warning lead time.
Restoration check	Test whether gap closure reduces C_E while F stabilizes/improves.
Falsification	Score failure if SR adds no value or frozen temporal ordering fails.
Replication	Repeat in a second healthcare dataset and then a heterogeneous domain.